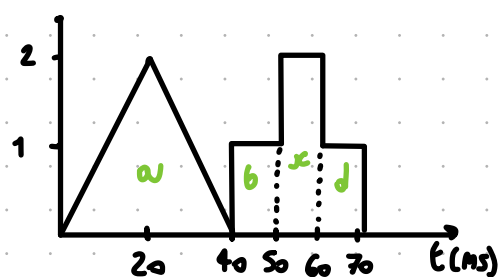


FOURIER SERIES



$$S_a(t) = 2 \text{TRIANG}\left(\frac{t-20}{40}\right) \longrightarrow C_n^a = 2 \cdot \frac{20}{2} \text{SINC}^2\left(f \frac{40}{2}\right) e^{-j2\pi f 20}$$

$$S_b(t) = 1 \text{RECT}\left(\frac{t-45}{10}\right) \longrightarrow C_n^b = 1 \cdot 10 \text{SINC}(f 10) e^{-j2\pi f 45}$$

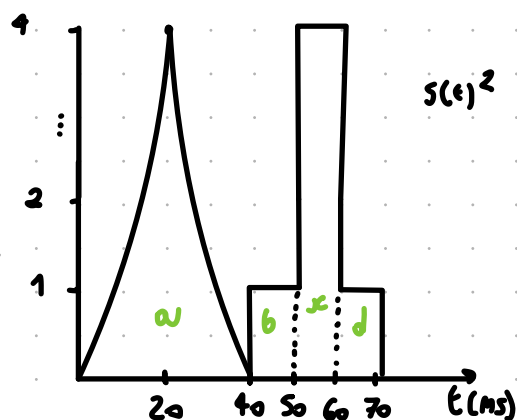
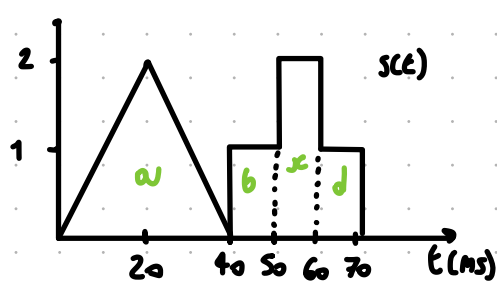
$$S_x(t) = 2 \text{RECT}\left(\frac{t-55}{10}\right) \longrightarrow C_n^x = 2 \cdot 10 \text{SINC}(f 10) e^{-j2\pi f 55}$$

$$S_d(t) = 1 \text{RECT}\left(\frac{t-65}{10}\right) \longrightarrow C_n^d = 1 \cdot 10 \text{SINC}(f 10) e^{-j2\pi f 65}$$

$$S(t) = S_a(t) + S_b(t) + S_x(t) + S_d(t)$$

SIGNAL POWER

$$P = \sum_{n=-\infty}^{\infty} |C_n|^2 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} s(t)^2 dt$$



(a) $y = ax^2$

$4 = a(20)^2 \rightarrow t: a \rightarrow \infty \rightarrow a = \frac{1}{100}$

$\int_0^{20} a t^2 dt = \frac{1}{100} \int_0^{20} t^2 dt = \frac{1}{100} \left[\frac{t^3}{3} \right]_0^{20} = 26.6$

$$P = \frac{1}{70} (2 \cdot 26.6 + 10 \cdot 1 + 10 \cdot 4 + 10 \cdot 1) \approx 1.61$$